Lazy ring + one directed shortcut: β sweep, tail mechanism, and jump-over extension report

Key conclusions and key figures

With $N = 100, q = 2/3, \rho = 1$ and only β varied, both peaks move to earlier times and the tail rate γ increases with β; at the same time, $K = 4$ sustains a wider bimodal window and a deeper valley. Figure 1 summarizes the peak and tail trends.

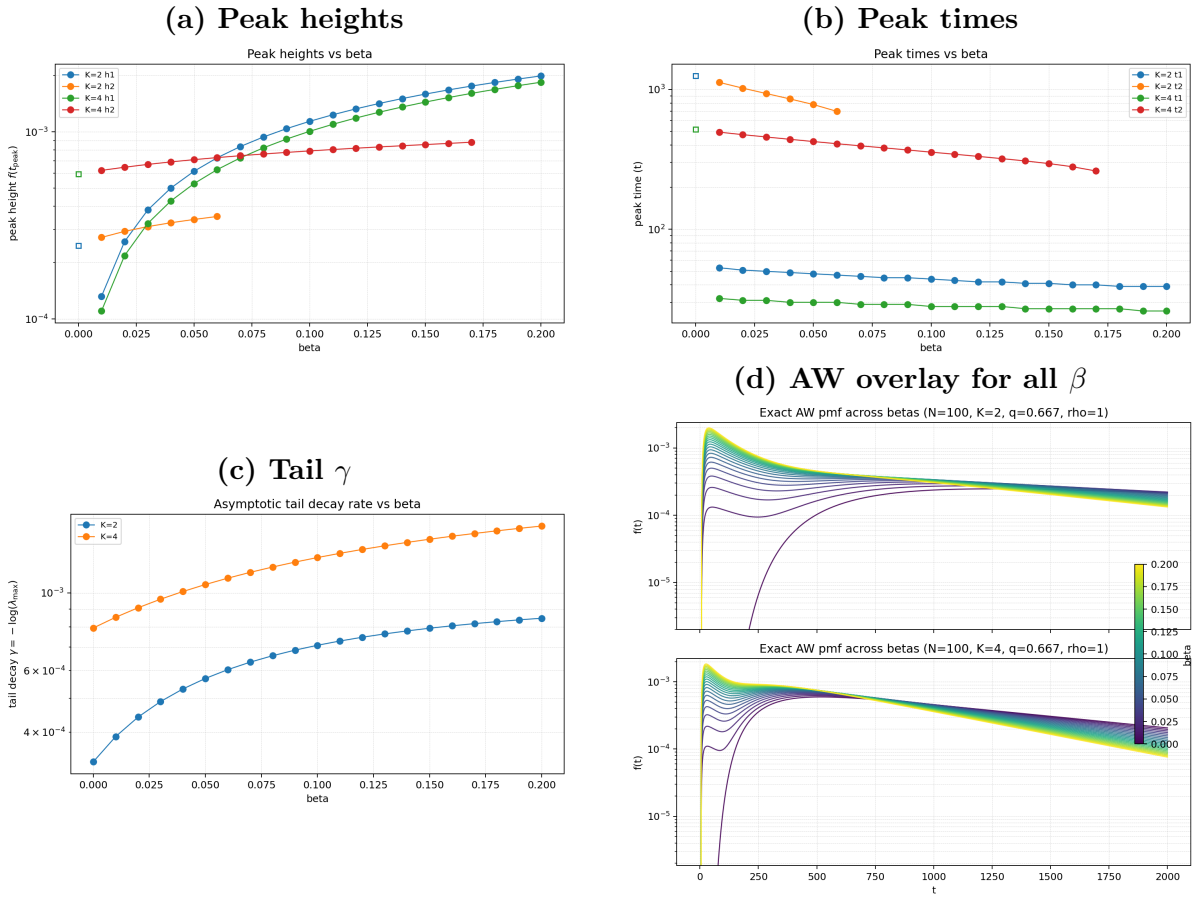

Figure 1: Exact AW results under a β-only sweep: peak heights, peak times, tail-decay rate γ, and the full-β overlay of $f(t)$; $\beta = 0$ is the no-shortcut baseline and is shown as a standalone marker.

At $\beta = 0.01$, the representative $f(t)$ highlights the valley contrast: `hv_over_max` ≈ 0.34 for $K = 2$ versus about 0.15 for $K = 4$, indicating a deeper valley and clearer separation for $K = 4$; at the same β, $K = 4$ has earlier (t_1, t_2) and larger γ, consistent with faster time scales and lighter tails. Figure 2 shows the peak–valley window directly.

The tail mechanism is governed by the transient spectral radius, $\gamma = -\log(\lambda_{\max})$, and $\log S(t)$ is close to linear in the tail (Fig. 3).

MC classification confirms that jump-over appears only for $K = 4$. We sort trajectories by whether the shortcut is used at least once ($C0$ vs $C1p$) and whether a jump-over occurs ($J0$ vs

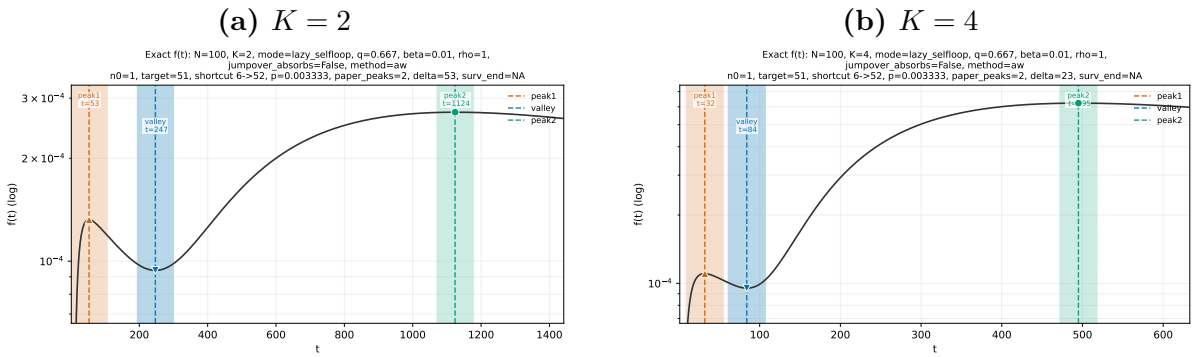

Figure 2: Representative exact $f(t)$ (AW), zoomed to the peak/valley window.

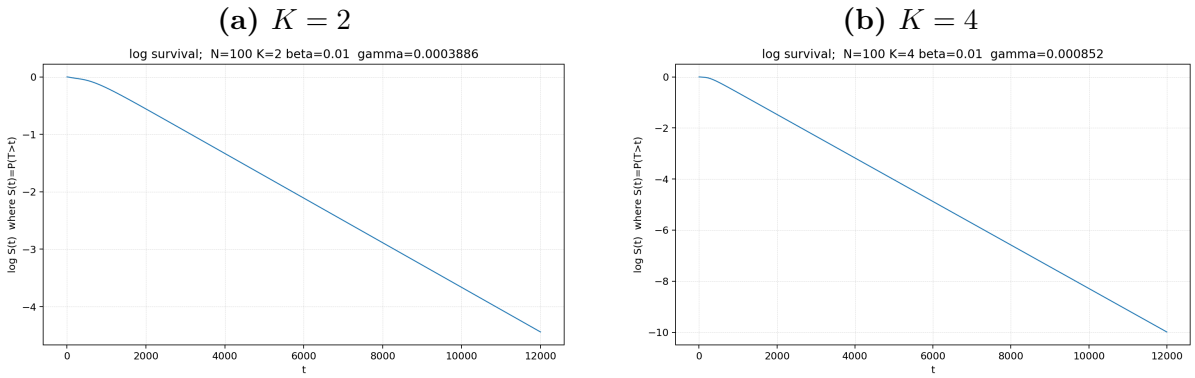

Figure 3: Log survival $\log S(t)$ at $\beta = 0.01$, consistent with the spectral-radius tail rate γ.

$J1p$), giving the four classes `C0J0`, `C1pJ0`, `C0J1p`, and `C1pJ1p`. Under this lens, $K = 2$ keeps J-related classes at zero across windows: peak1 is essentially all `C1pJ0`, the valley is mostly `C1pJ0` with a small `C0J0` tail, and peak2 is dominated by `C0J0` with a smaller `C1pJ0` share. For $K = 4$, peak1 splits into roughly two-thirds `C1pJ0` and one-third `C1pJ1p`; the valley shows a similar one-third $J1p$ share, while peak2 flips to `C0`-dominated with about a quarter `C0J1p`. These $J1p$ paths correspond to delayed crossings without immediate absorption, and the curves vary only mildly with β (Fig. 4).

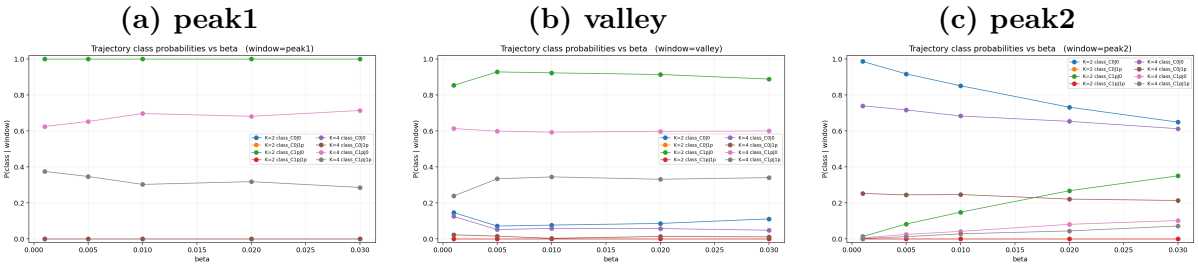

Figure 4: Conditional class probabilities vs β for each window ($N = 100$).

To choose a representative bimodal setting, we apply $t_2/t_1 \geq 10$, `hv_over_max` ≤ 0.6, and `h2_over_h1` ≥ 0.05, selecting the smallest β that satisfies both $K = 2$ and $K = 4$; the scan yields $\beta_* = 0.01$.

Fixing $\beta_* = 0.01$ and sweeping N shows t_2 shifting right, h_2 decreasing, and γ decreasing (heavier tails), while t_1 stays roughly stable; see Figures 5 and 6.

The MC sweep over N sharpens the same contrast. For $K = 2$, everything remains in $J0$: peak1 is locked to `C1pJ0`, the valley is mostly `C1pJ0`, and peak2 mixes `C0J0` with `C1pJ0` as N changes. For $K = 4$, once the valley/second peak are well formed, peak1 and the valley hold a stable ∼one-third $J1p$ share (largely `C1pJ1p`), while peak2 stays `C0`-dominated with about a quarter `C0J1p`, showing that jump-over contributions persist across N (Fig. 7).

This extension builds on the main jump-over report (lazy_jump) and uses the same model

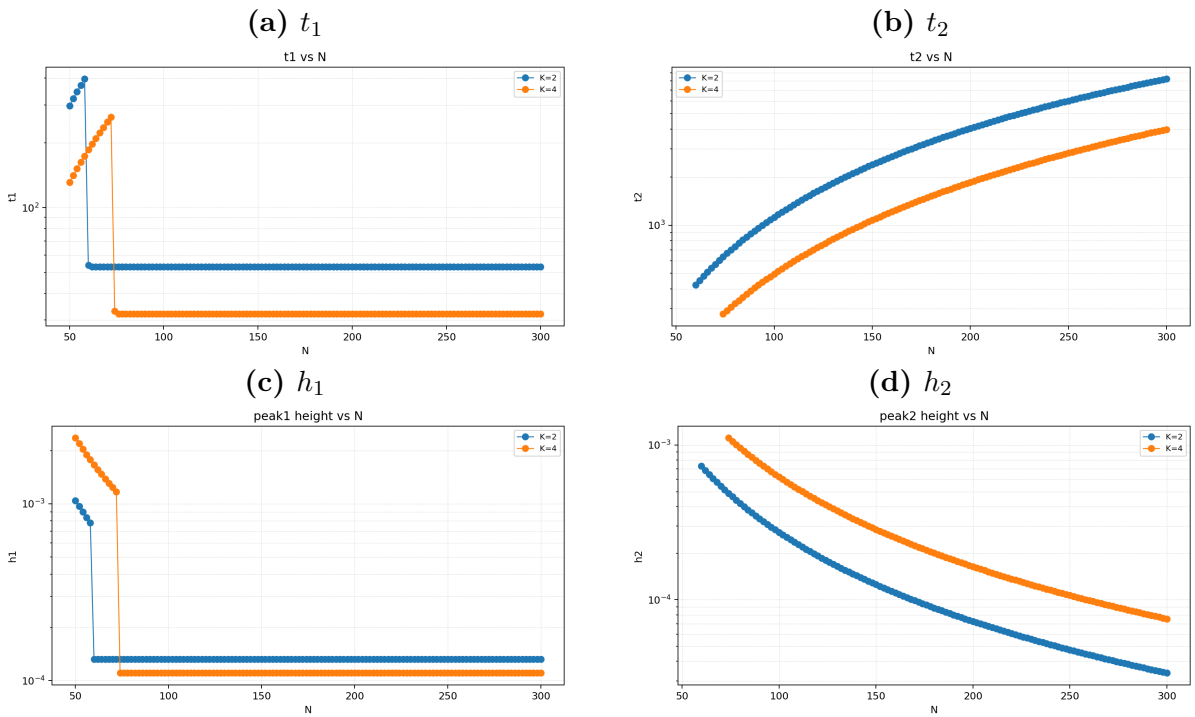

Figure 5: Peak locations and heights vs N at $\beta = 0.01$.

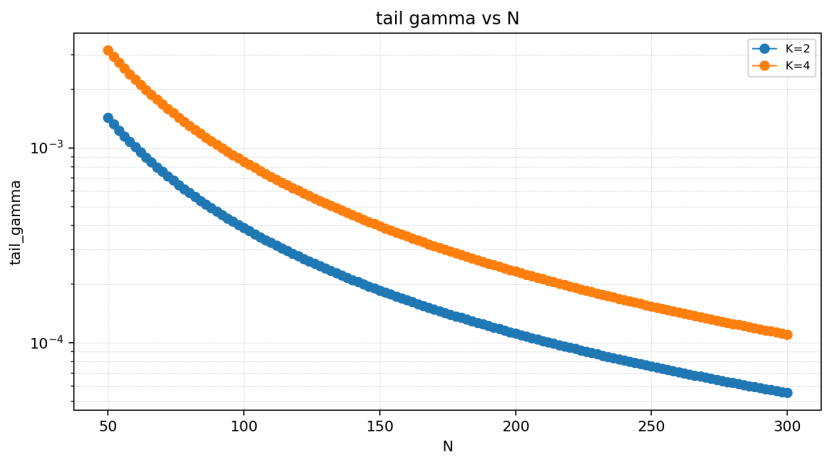

Figure 6: Tail decay rate γ vs N at $\beta = 0.01$.

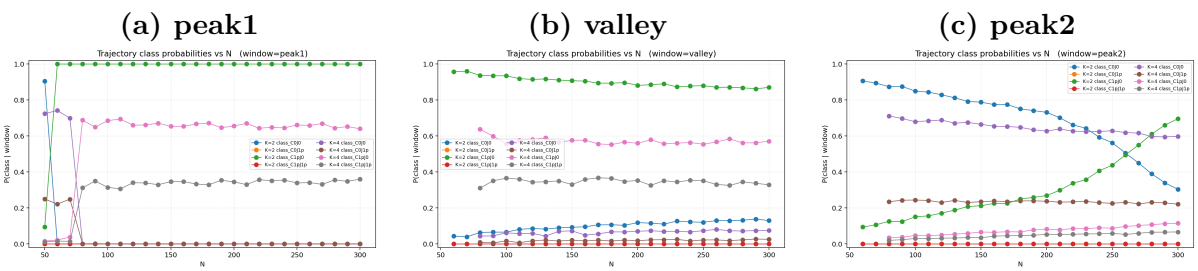

Figure 7: Class proportions vs N at fixed $\beta = 0.01$.

and notation. The remainder gives the theory and valley definition, then the β sweep, MC classes, N sweep, MC-N, and reproducibility.

1 Theory: generating functions, AW inversion, and tail decay

This section provides the theoretical backbone for our exact pmf computation and tail diagnostics. We consider a lazy random walk on an N-node ring with $K = 2k$ neighbors: with probability q the walker moves on the ring, choosing uniformly from steps $\pm 1, \ldots, \pm k$, and with probability $1 - q$ it stays put. A directed shortcut from src to dst is added.

1.1 Spectral decomposition and the resolvent without a defect

Let P_0 denote the defect-free transition matrix. For non-source nodes, the one-step transition is

$$P_0(i \to i) = 1 - q, \qquad P_0(i \to i \pm r) = \frac{q}{K}, \quad r = 1, \ldots, k.$$

Because P_0 is translation-invariant (a circulant matrix), its eigenvectors are Fourier modes

$$\phi_\ell(x) = \exp\left(\frac{2\pi i \ell x}{N}\right), \qquad \ell = 0, \ldots, N-1,$$

with eigenvalues

$$\lambda_\ell = (1-q) + \frac{q}{K}\sum_{r=1}^{k}\left(e^{2\pi i \ell r/N} + e^{-2\pi i \ell r/N}\right) = (1-q) + \frac{2q}{K}\sum_{r=1}^{k}\cos\left(\frac{2\pi \ell r}{N}\right). \tag{1}$$

Define the resolvent (matrix generating function)

$$S^{(0)}(z) = (I - zP_0)^{-1}, \qquad |z| < 1.$$

By spectral decomposition,

$$S^{(0)}(z)_{x,y} = \frac{1}{N}\sum_{\ell=0}^{N-1}\frac{\exp\left(\frac{2\pi i \ell (x-y)}{N}\right)}{1 - z\lambda_\ell}. \tag{2}$$

One may further define a displacement form $Q(d, z) := S^{(0)}(z)_{y+d,y}$ to efficiently assemble defect quantities.

1.2 Directed shortcut under the self-loop rule: a rank-one update

In the `lazy_selfloop` mode, the shortcut probability is

$$p = \beta(1-q),$$

and the same amount is removed from the self-loop probability at the source: $P(\text{src} \to \text{src}) = (1-q) - p$ and $P(\text{src} \to \text{dst}) = p$. Hence the transition matrix is a rank-one perturbation of P_0:

$$P = P_0 + p\, e_{\text{src}}(e_{\text{dst}} - e_{\text{src}})^\top, \tag{3}$$

where e_i is the standard basis vector. Therefore

$$I - zP = (I - zP_0) - zp\, e_{\text{src}}(e_{\text{dst}} - e_{\text{src}})^\top.$$

Applying the Sherman–Morrison formula

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u},$$

one can obtain the required entries of the defect resolvent $S(z)$ from $S^{(0)}(z)$ with $O(1)$ additional scalar operations (as implemented in the repository pipeline).

1.3 First absorption generating function under geometric absorption (ρ)

Let tar be the target. Upon each visit to tar, absorption occurs with probability ρ, otherwise the process continues with probability $1-\rho$. Let

$$S_{i,j}(z) = \sum_{t\geq 0} \mathbb{P}(X_t = i \mid X_0 = j) z^t$$

be the visit generating function of the non-absorbed walk. Then the generating function of the first absorption pmf $f(t) = \mathbb{P}(T = t)$ for $t \geq 1$ is

$$\tilde{f}(z) = \sum_{t\geq 1} f(t) z^t = \frac{\rho\, S_{\mathrm{tar},n_0}(z)}{(1-\rho) + \rho\, S_{\mathrm{tar,tar}}(z)}. \tag{4}$$

For $\rho = 1$, this reduces to

$$\tilde{f}(z) = \frac{S_{\mathrm{tar},n_0}(z)}{S_{\mathrm{tar,tar}}(z)}.$$

1.4 Abate–Whitt (AW) inversion via FFT

Coefficient extraction can be discretized on a circle. With $z_m = re^{2\pi i m/L}$ $(r < 1)$,

$$f(t) \approx r^{-t} \frac{1}{L} \sum_{m=0}^{L-1} \tilde{f}(z_m) e^{-2\pi i m t/L}. \tag{5}$$

The right-hand side is an inverse DFT, thus FFT yields all $f(t)$ for $t = 1, \ldots, T_{\max}$ in $O(L \log L)$ time. We follow the repository's AW parameter selection and numerical stabilization.

1.5 Exponential tail decay and the spectral-radius rate γ

Let Q be the transient submatrix of the absorption chain. The survival function is

$$S(t) = \mathbb{P}(T > t) = \mathbf{1}^\top Q^t e_{n_0}.$$

By Perron–Frobenius theory for nonnegative matrices, there exists $\lambda_{\max} \in (0,1)$ such that

$$S(t) \sim c\, \lambda_{\max}^t \quad (t \to \infty).$$

We define the asymptotic exponential tail decay rate

$$\gamma = -\log(\lambda_{\max}), \tag{6}$$

which provides a stable and comparable tail metric across β, N, and K. Numerically, we construct Q and compute its spectral radius; for $\rho = 1$ we remove the target state, and for $\rho < 1$ we multiply the target row by $(1-\rho)$ to represent absorption leakage.

2 Comparable definition of the valley and windowing

Let $f(t) = \mathbb{P}(T = t)$ for $t \in \mathbb{N}^+$ denote the pmf of the first absorption time. We detect candidate peaks as strict local maxima:

$$f(t) > f(t-1), \qquad f(t) > f(t+1), \qquad f(t) \geq h_{\min}. \tag{7}$$

We further filter secondary peaks by a relative-height threshold: $f(t) \geq$ `second_rel_height` $\cdot \max_u f(u)$.

2.1 V0: Default valley (minimum between the two peaks)

Given the first two peaks in chronological order $t_1 < t_2$, define

$$t_v = \arg\min_{t_1<t<t_2} f(t). \tag{8}$$

This definition is parameter-light and guarantees that t_v lies strictly between the two peaks, facilitating comparisons across β and across K.

2.2 V1: Relative-distance window scale (for trajectory-type comparisons)

Let $\Delta T = t_2 - t_1$ be the peak separation and define the half window width

$$\Delta = \max\left(1, \lfloor \delta_{\text{frac}} \Delta T \rfloor\right). \tag{9}$$

Each sample is assigned to the nearest center in $\{t_1, t_v, t_2\}$ if $\min_{c\in\{t_1,t_v,t_2\}} |T-c| \le \Delta$, otherwise it is assigned to `other`. Because Δ scales proportionally with ΔT, the windowing remains comparable across β.

2.3 V2: Robust valley (excluding regions close to peaks)

To avoid valleys too close to a peak (which may happen when peaks are close), we may define

$$t_v = \arg\min_{t_1+\alpha\Delta T \le t \le t_2-\alpha\Delta T} f(t), \tag{10}$$

with $\alpha \in (0, 1/2)$ (e.g., $\alpha = 0.1$). We use V2 as a sensitivity-analysis convention.

3 Beta-only sweep at fixed N ($N = 100$)

We fix $N = 100$, $q = 2/3$, $\rho = 1$, and the `lazy_selfloop` convention, and vary only β. Figure 1 summarizes exact AW peak heights/locations and the tail-decay metric, together with an all-β overlay of $f(t)$.

Peak–valley quantification We record the valley depth `hv_over_max` $= h_v/\max(h_1, h_2)$ and `h2_over_h1` (see CSV). Comparing $K = 2$ and $K = 4$ highlights the depth difference in the valley. At $\beta = 0.01$, $K = 2$ has `hv_over_max` ≈ 0.34, while $K = 4$ yields ≈ 0.15, indicating a deeper valley and clearer separation for $K = 4$. At the same β, $K = 4$ shows earlier t_1, t_2 and larger γ, consistent with faster time scales and lighter tails. Note that $K = 2$ has $J \equiv 0$ (no jump-over), while $K = 4$ allows $J > 0$, which will reshape class compositions in the MC sweep below.

Representative $f(t)$ with improved readability Figure 2 shows $\beta = 0.01$ with the time axis focused on the peak–valley interval (avoiding the long vertical stretch near $t \approx 0$), making the peak/valley contrast visually clear.

Tail decay diagnostics The asymptotic tail is controlled by $\gamma = -\log(\lambda_{\max})$. Figure 3 confirms the near-linear decay of $\log S(t)$. At $\beta = 0.01$, $K = 2$ has $\gamma \approx 3.89 \times 10^{-4}$ while $K = 4$ has $\gamma \approx 8.52 \times 10^{-4}$, matching the γ–vs–β curves.

3.1 Automatic choice of β

We apply the same thresholds ($t_2/t_1 \ge 10$, `hv_over_max` ≤ 0.6, `h2_over_h1` ≥ 0.05) and choose the smallest β that satisfies them for both $K = 2$ and $K = 4$. The scan yields $\beta_* = 0.01$ (see `outputs/selected_beta.txt`).

These exact AW results set the stage for the MC class analysis in the next section.

4 MC sweep: four classes vs β ($N = 100$)

We use MC to estimate class probabilities within each window, defined by (t_1, t_v, t_2) and $\Delta = \lfloor \delta_{\text{frac}}(t_2 - t_1) \rfloor$. The four mutually exclusive classes are: `class_C0J0` (no shortcut, no jump-over), `class_C1pJ0` (shortcut, no jump-over), `class_C0J1p` (no shortcut, jump-over), `class_C1pJ1p` (shortcut and jump-over). For $K = 2$ we have $J \equiv 0$, so J-related classes should be 0 or nearly 0. Figure 4 reports the conditional class probabilities across windows as β varies.

Interpretation highlights For $K = 2$, J-related classes vanish, confirming that jump-over is absent. For $K = 4$, the valley and second-peak windows show larger shares of J-related classes, consistent with trajectories that cross the target without absorbing and hence contribute to delayed arrivals.

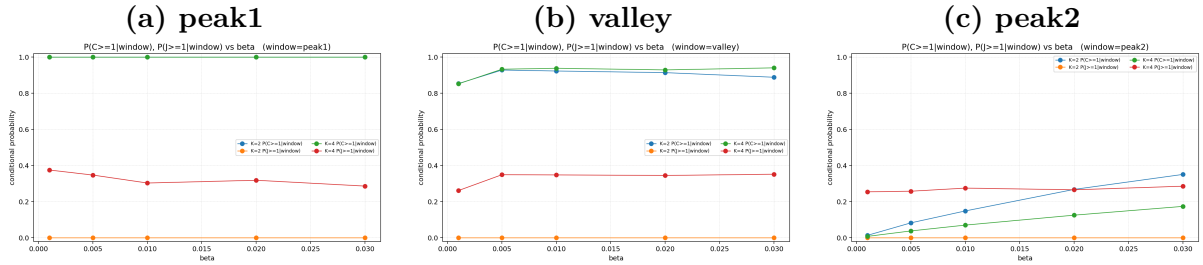

Figure 8: $\mathbb{P}(C \geq 1 \mid \text{window})$ and $\mathbb{P}(J \geq 1 \mid \text{window})$ vs β.

With the β-direction class structure established, we now fix β_* and examine how the slow time scale varies with N.

5 Fix $\beta = 0.01$ and sweep N (exact AW)

Using the automatically selected $\beta_* = 0.01$ and fixing $q = 2/3$, $\rho = 1$, and `lazy_selfloop`, we sweep even N and track peak locations/heights and tail decay. Figure 5 shows that t_2 shifts right as N grows (a slower second time scale), while h_2 decreases; the tail rate γ decreases with N, indicating a heavier tail, and the trend is summarized in Figure 6.

The next section complements this with MC class statistics across N.

6 MC sweep: four classes vs N (fixed $\beta = 0.01$)

With $\beta = 0.01$ fixed, we scan even N and estimate class proportions within each window. Recall: C records whether the shortcut is used at least once ($C0$ vs $C1p$) and J records whether a jump-over occurs ($J0$ vs $J1p$), so the four classes are `C0J0`, `C1pJ0`, `C0J1p`, and `C1pJ1p`. Figure 7 summarizes the class proportions vs N.

Comparison highlights For $K = 2$, J-related classes are essentially zero; for $K = 4$, J-related classes appear prominently in the valley and second-peak windows, reflecting jump-over contributions to delayed absorption.

See the reproducibility section for the exact commands and parameters.

7 Reproducibility (commands and outputs)

Run the following inside `reports/lazy_jump_ext/`.

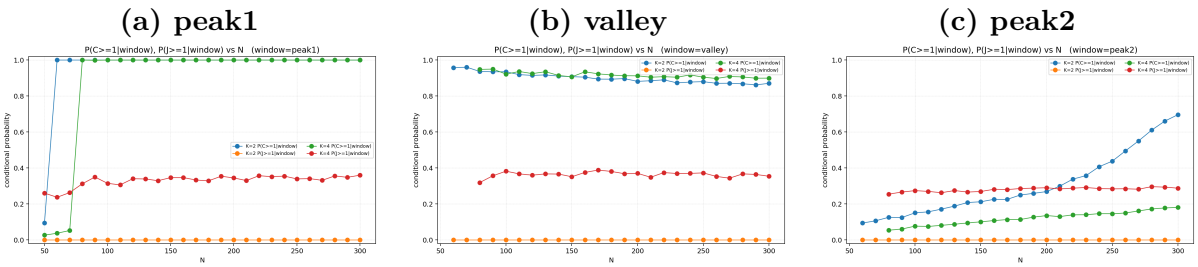

Figure 9: $\mathbb{P}(C \geq 1 \mid \text{window})$ and $\mathbb{P}(J \geq 1 \mid \text{window})$ vs N.

1. β sweep (exact AW)

```
python code/beta_sweep_peaks_tail.py \
  --N 100 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --beta-min 0.0 --beta-max 0.2 --beta-num 21 \
  --outdir outputs/beta_sweep_N100
```

Outputs: `outputs/beta_sweep_N100/beta_sweep_metrics_N100.csv` and the three plots.

2. Automatic β selection

```
python code/choose_beta.py \
  --csv outputs/beta_sweep_N100/beta_sweep_metrics_N100.csv \
  --Ks 2 4 --macro_ratio 10 --hv_over_max 0.6 --min_h2_over_h1 0.05 \
  --fallback_beta 0.02 \
  --out outputs/selected_beta.txt
```

3. Fix β and sweep N (exact AW)

```
python code/N_sweep_peaks_tail.py \
  --K 2 4 --beta 0.01 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --N-min 50 --N-max 300 --N-step 2 --only-even \
  --outdir outputs/N_sweep_beta_selected
```

4. MC sweep over β (four classes)

```
python code/mc_sweep_beta_classes.py \
  --N 100 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --betas 0.001 0.005 0.01 0.02 0.03 \
  --n-walkers 200000 --seed 0 \
  --outdir outputs/mc_beta_sweep_N100
```

5. MC sweep over N (fixed β)

```
python code/mc_sweep_N_classes.py \
  --beta 0.01 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --N-min 50 --N-max 300 --N-step 10 --only-even \
  --n-walkers 200000 --seed 0 \
  --outdir outputs/mc_N_sweep_beta_selected
```

6. Tail diagnostics ($\log S(t)$)

```
python code/tail_diagnostics.py \
  --N 100 --K 2 --beta 0.01 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --max-steps-aw 12000 --outdir outputs/tail_diag
python code/tail_diagnostics.py \
  --N 100 --K 4 --beta 0.01 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --max-steps-aw 12000 --outdir outputs/tail_diag
```

7. Overlay all-β exact curves (optional)

```
python code/plot_aw_all_betas.py \
  --N 100 --Ks 2 4 --q 0.6666666667 --rho 1.0 --mode lazy_selfloop \
  --beta-min 0.0 --beta-max 0.2 --beta-num 21 \
  --out outputs/beta_sweep_N100/aw_all_betas_N100.png
```

8. Build the extension PDFs

```
latexmk -xelatex -interaction=nonstopmode ring_lazy_jump_ext_cn.tex
latexmk -pdf -interaction=nonstopmode ring_lazy_jump_ext_en.tex
```